CARMEL VIDYA BHAVAN TRUST'S

CHRIST COLLEGE PUNE

(Affiliated to Savitribai Phule Pune University)



Criterion VII

7.1.2

- Alternate Sources of Energy & Energy Conservation Methods
- Management of Degradable & Non-degradable Waste
- Water Conservation
- Green Campus Initiatives
- Disabled-friendly, barrier free environment



FACILITY & INITIATIVES

7.1.2 RELEVANT DOCUMENTS

7.1.2 Alternative sources of energy and energy conservation measures

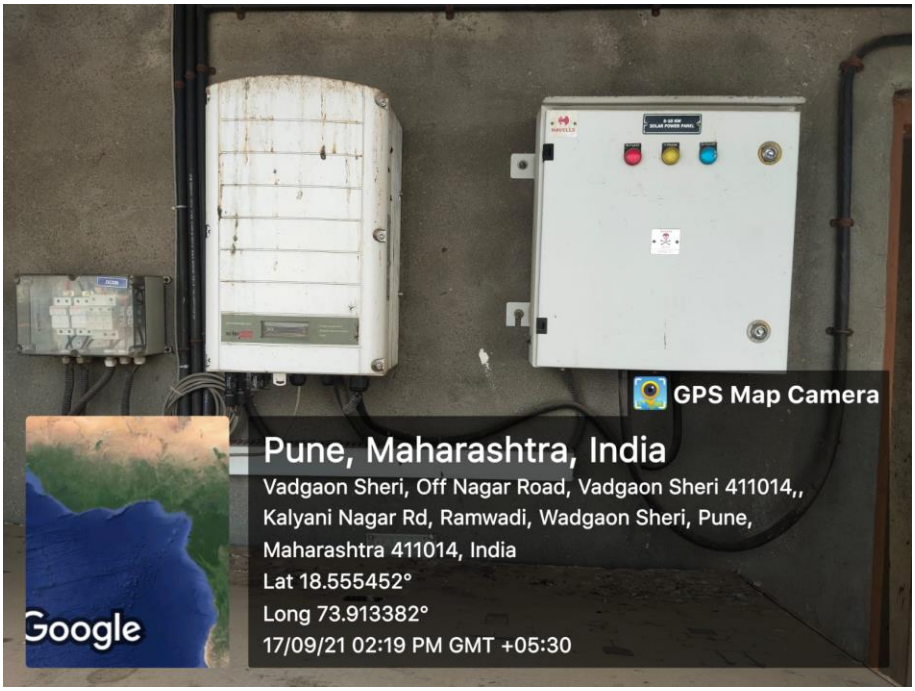
Conserving energy produces a higher quality of life. In addition, it helps to create a healthier planet, or at least helps to sustain the resources we already have. Institute uses following alternative sources of energy and energy conservation measures:

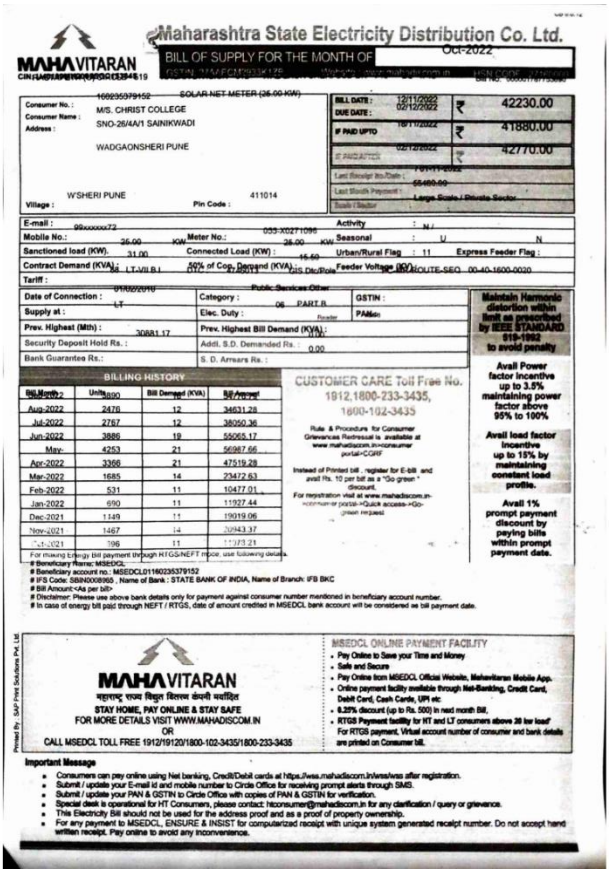
1. SOLAR ENERGY

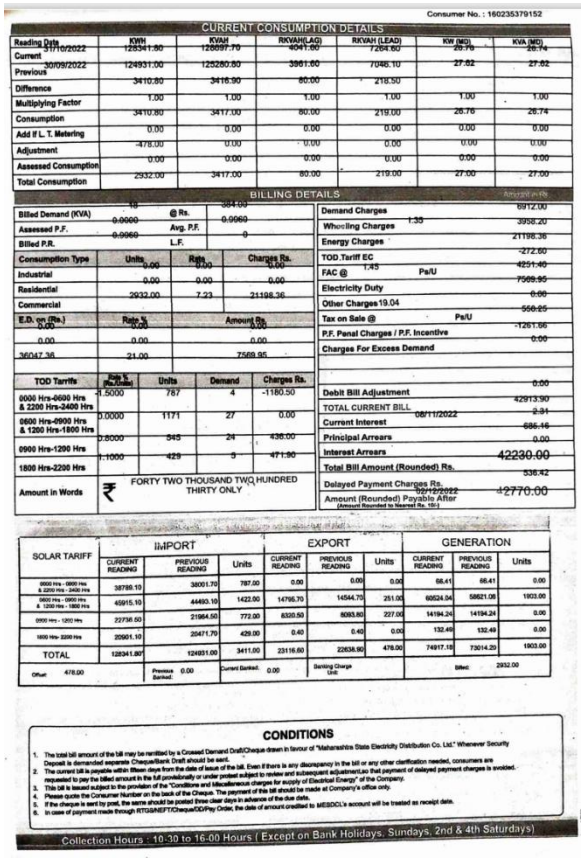
The institution has installed solar panels on the roof of the college building that produces renewable energy to try to meet the increased electricity demand. Currently, the institute has solar panels that contribute to 37% of the overall monthly electricity consumption. Solar Panels of 20.16 kVA are installed on the college building.



2. WHEELING TO THE GRID



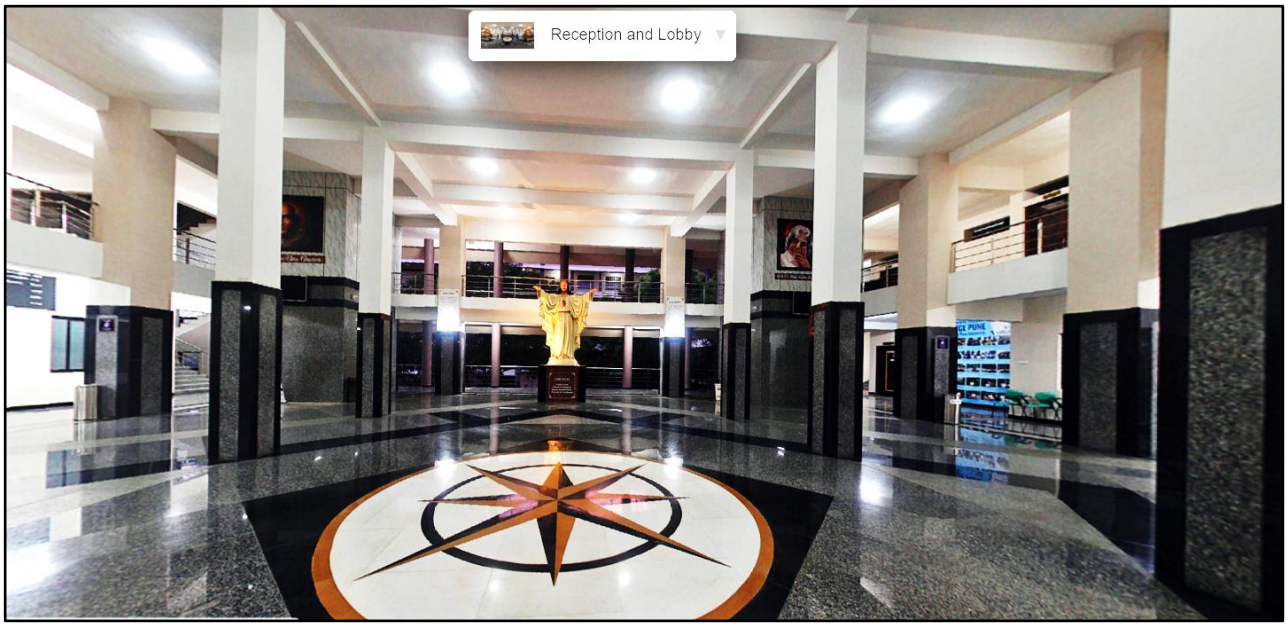


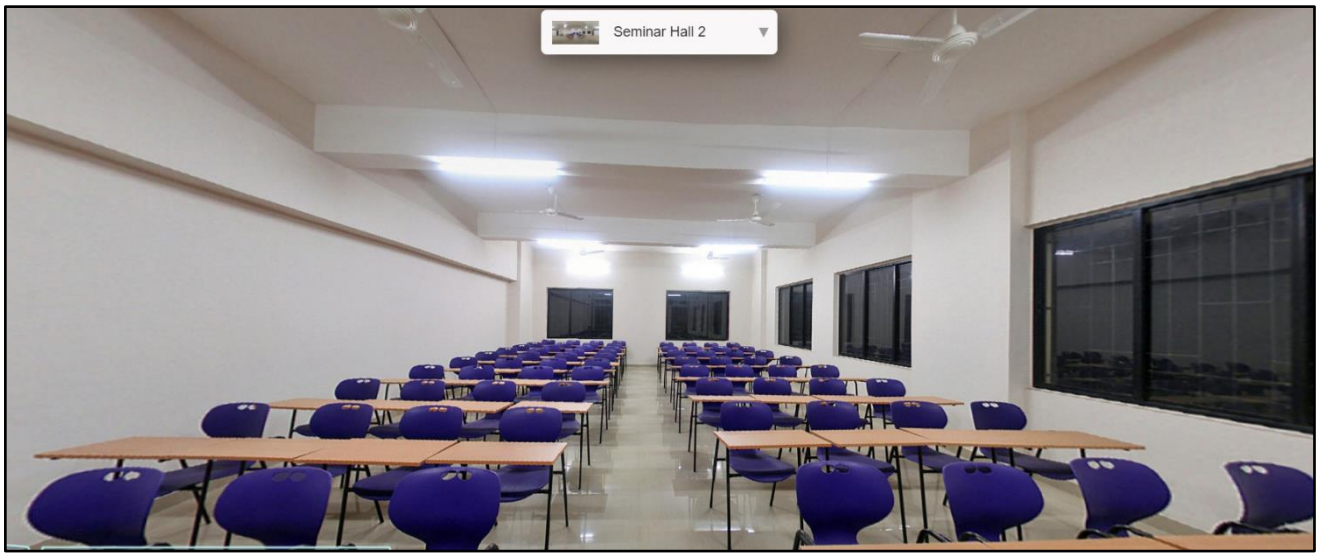


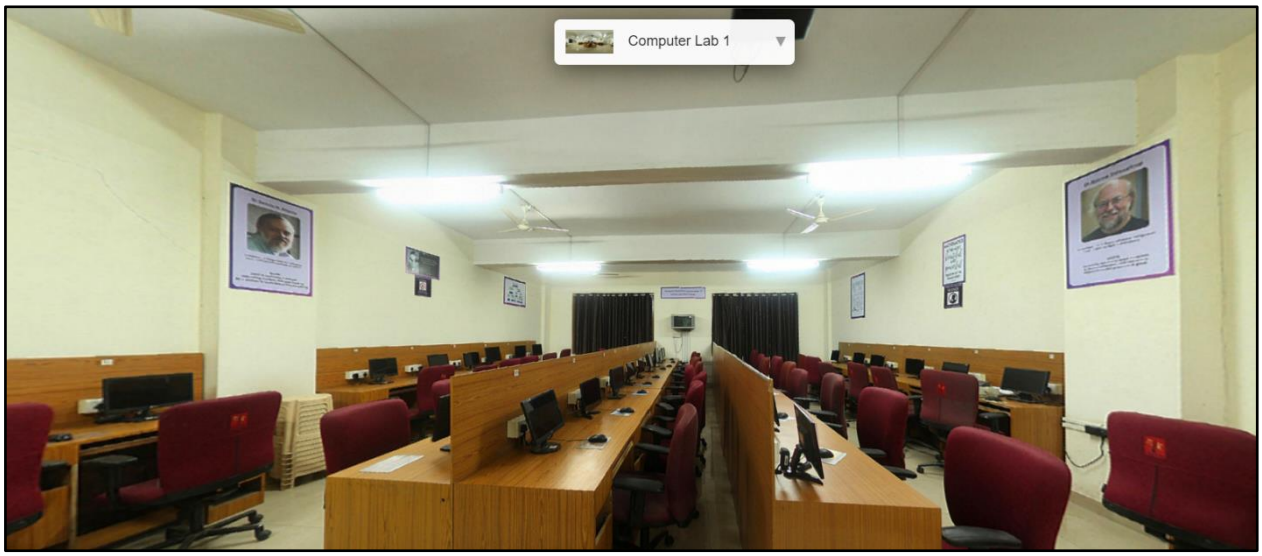
3. USE OF LED BULBS/ POWER EFFICIENT EQUIPMENT

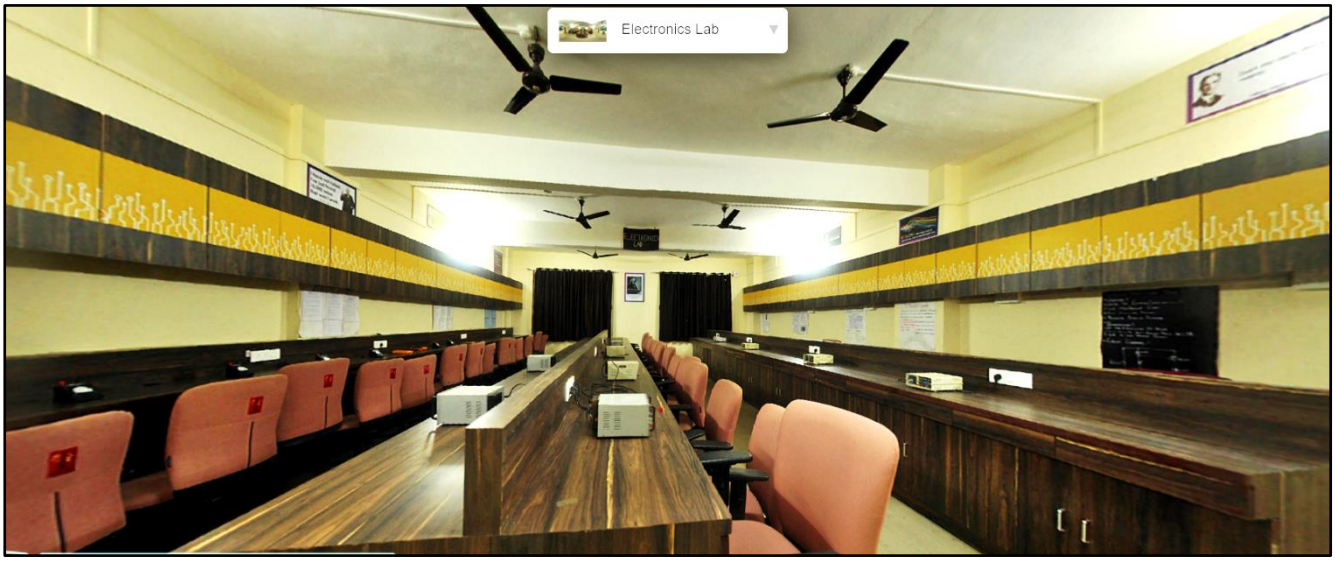
Around 22% power saving in the campus is done by using and replacing the conventional light bulbs by LEDs.

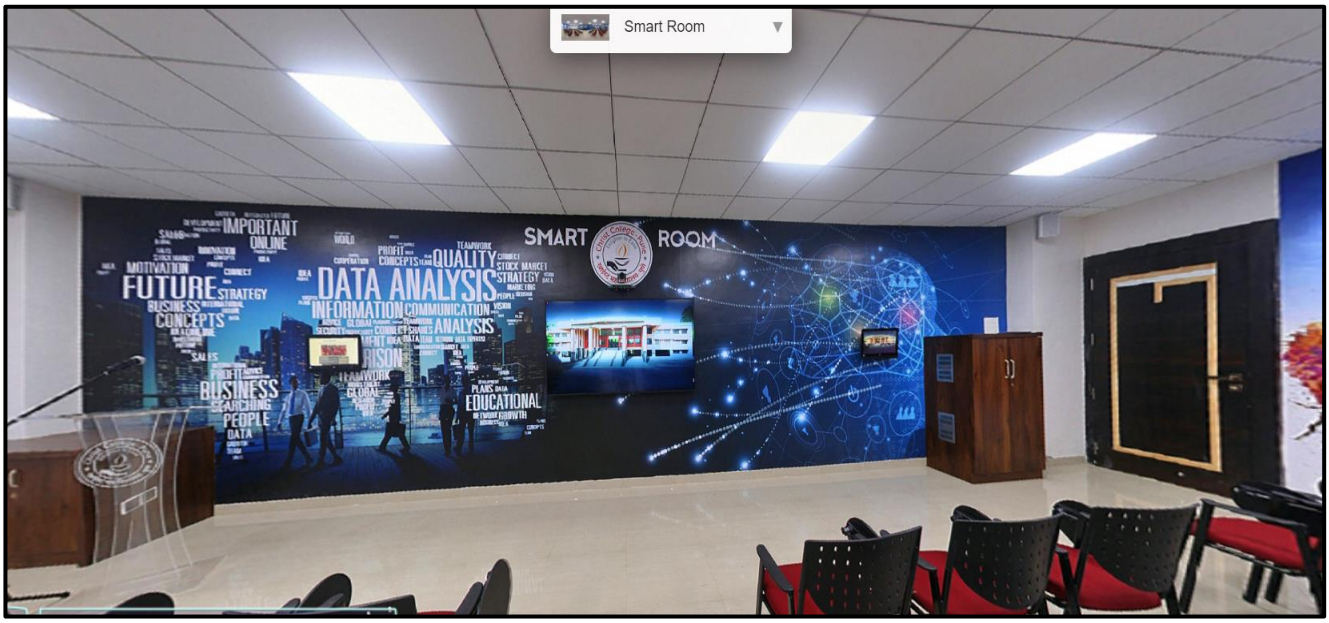
Use of LEDs throughout the Campus

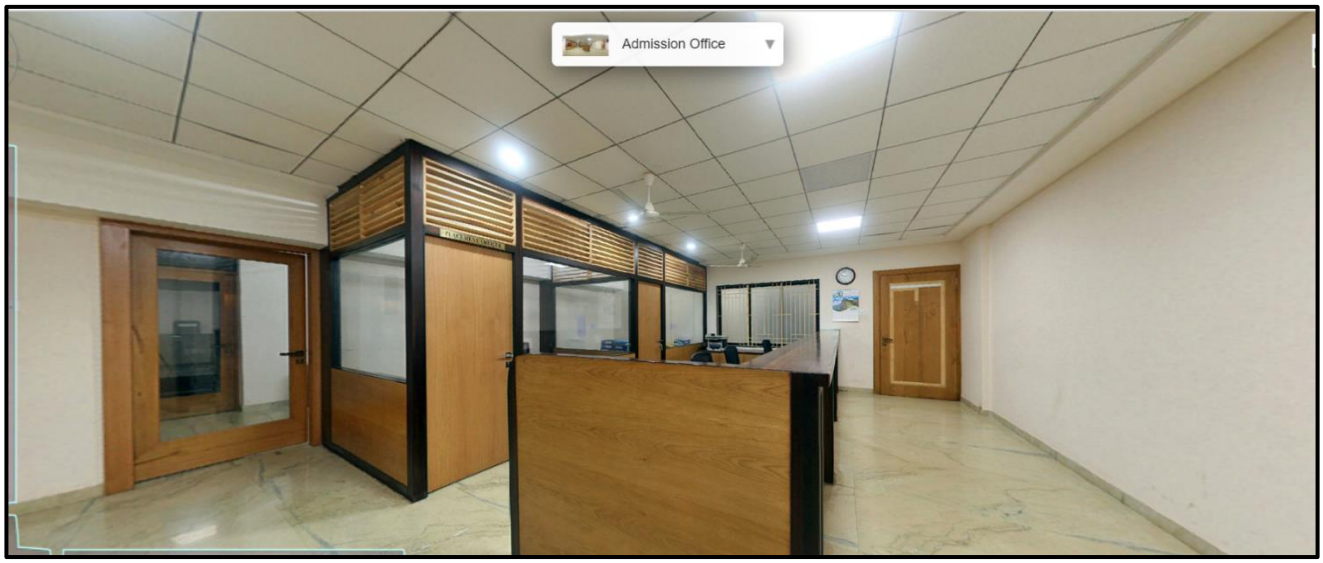





Computer Lab 1


Electronics Lab


Smart Room
SMART ROOM
DATA ANALYSIS
FUTURE
BUSINESS
EDUCATIONAL


Admission Office

7.1.2 Management of various types of degradable and nondegradable waste

Solid waste management

- Color coded bins for dry, wet, plastic and sanitary waste are used for segregation.
- The students are accustomed to use bins placed in each classroom and all around the campus.
- Vermiculture compost unit is constructed to convert the organic waste into fertilizer.
- Composting pit is used to convert green waste into organic manure for maintaining campus garden.

Liquid waste management

- Recycling of water is promoted through rain water harvesting. 5Lakh liters capacity tank is constructed to recycle and save water which is utilized for construction work, washing purposes and also provided to the local community during scarcity.
- Waste water of washrooms is recycled for gardening purpose.

Bio-medical waste management

- The biomedical waste- sanitary pads are disposed through napkin incinerator installed in campus.

E-waste management

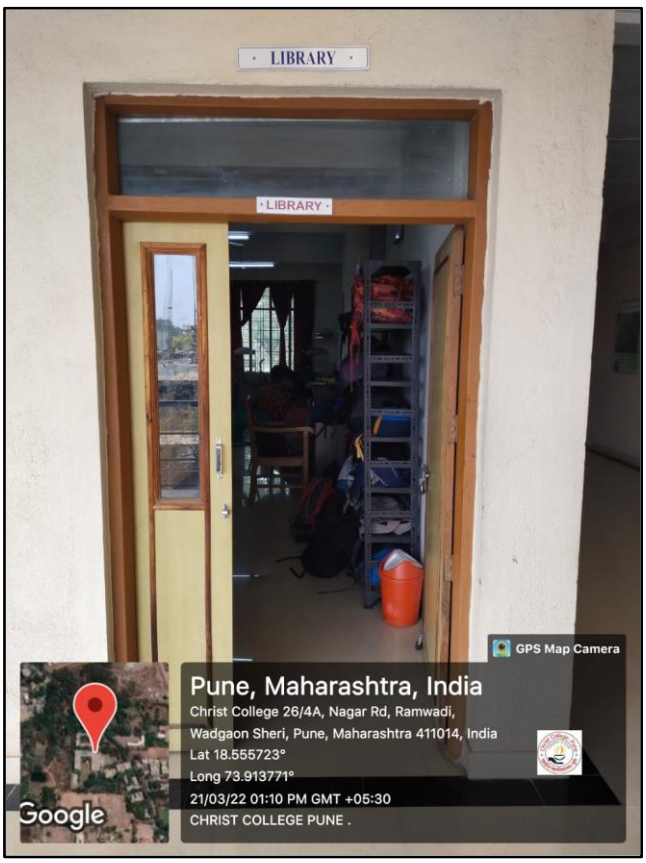
- E-waste disposal is stored in a warehouse from where it is handed over to e-waste collectors.

1. SOLID WASTE MANAGEMENT





Segregated Dry & Wet waste bins in campus





Vermicomposting unit



Composting pit

2. LIQUID WASTE MANAGEMENT



Liquid waste management through Roof top Rain water harvesting system



Recycled water used for Gardening

3. BIO-MEDICAL- SANITARY WASTE





Vending machine and Incinerator installed in campus

4. E-WASTE MANAGEMENT





E-Waste Management Drive

As we stand in the midst of extreme environmental and climate deterioration. E-waste hazard is one of the major issues we face in this digital era. Electronic waste or e-waste describes discarded electrical or electronic devices. For example old mobile phones, chargers, laptops keypads, pen drives etc.

To curb this issue the NSS UNIT of Christ College, Pune organised an E-waste awareness drive for a week starting from 23rd of May till 28th of May. All the staff and students were requested to contribute to this drive by bringing any electronic waste and discarding it in e-waste dustbin placed in the lobby area near the college office. This e-waste was then recycled by the Recycling Company (Threco Recycling LLP). A certificate was issued by the company for receiving 318.9 Kgs of E-waste.








Pune, Maharashtra, India




Pune, Maharashtra, India

















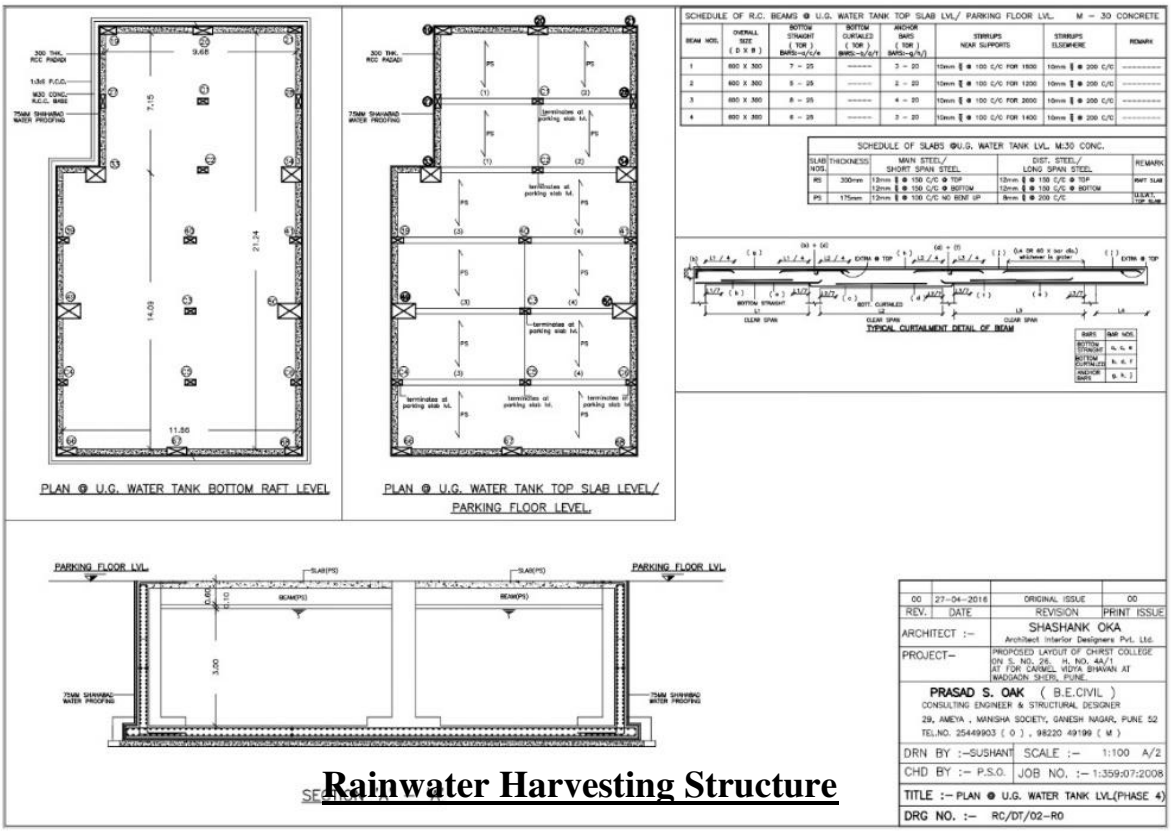
3. Water conservation facilities available in the Institution: Maintenance of water bodies and distribution system in the campus.

A significant portion of the College's original terrain has been preserved, which greatly facilitates groundwater recharge and water percolation. The campus's beautiful ambiance, intellectual setting, and green spaces with trees and lawns, wide hallways, and expansive playgrounds inspire students and staff to learn, develop their talents, and foster the teaching-learning process.

The college practices water conservation through:

1. RAIN WATER HARVESTING

The College is well equipped with the facility of rain water harvesting. We are using roof top type rain water harvesting system which collects rain water from the roof of the building and divert it to a storage tank with a capacity of 502830 liters. The dimension of the tank is 13.84m (length) x 11.72m (width) x 3.10m (height). The recycled water is used to cater the need for construction work, washing purposes, irrigating the lawn and landscape, flush water in toilets, as fire extinguisher etc. During water crisis, this stored water is used by the local community too.



Rainwater Harvesting Structure

2. OPEN WELL RECHARGE

A huge open well, constructed in the campus is recharged by rain water. The water is used for washing and irrigation purposes.



Open Well in campus

3. CONSTRUCTION OF TANKS

As the water crisis continues to become severe, there is a need to save it properly. The college has rooftop tanks of around 20,000 Lts. and 5000 Lts. capacity. Water

from corporation is stored in the underground water tank in the campus. Separate tank of 502830 Lts. is constructed to collect rain water.





Rooftop and underground water tanks

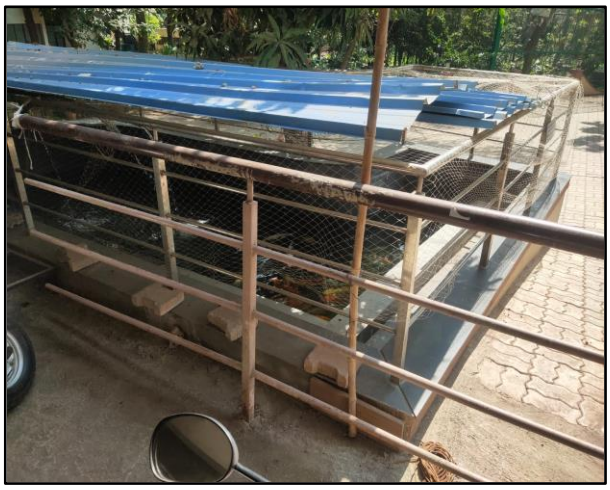


Water tank for birds and watering in vegetable garden

4. WASTE WATER RECYCLING

Wastewater discharged as effluent from washroom, canteen wastewater and other are used for gardening, watering of trees, maintaining fish tank etc.

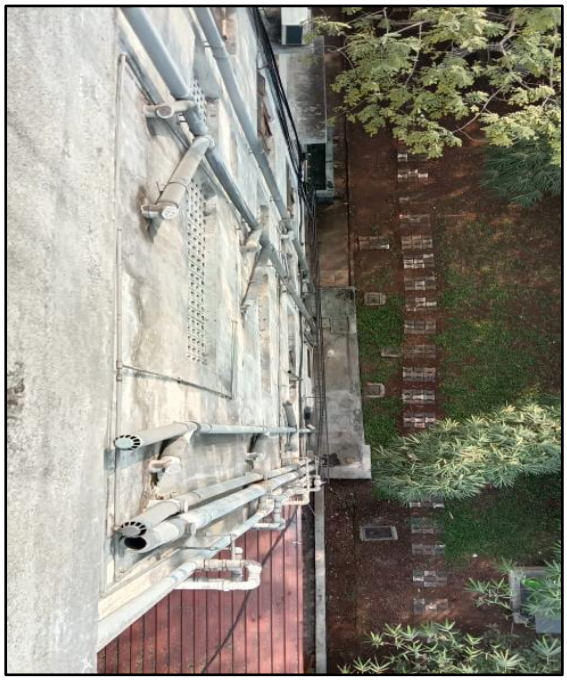




5.MAINTENANCE OF WATER BODIES AND DISTRIBUTION SYSTEM IN THE CAMPUS

The ground water is pumped into storage tanks located at different places in the campus. The water is distributed through well laid pipe network. Drinking water from Municipal corporation is supplied through a separate set of distribution pipes. Drainage pipes are connected with Municipal Corporation drainage system. Entire distribution system is well supervised to ensure that there are no leakages and wastage through joints, valves, overflow etc. Water Coolers and Water Filters are provided separately for staff and students. Staff, students and stakeholders of the college are made well aware to use water economically and efficiently.





Drainage lines connected to corporation drainage lines



Water Filters & coolers installed in campus

4. GREEN CAMPUS INITIATIVES

A Green Campus is a place where environmental-friendly practices and education combine to promote sustainable and eco-friendly environment in the campus. The institute has a serene, lush green campus providing a natural environment for the holistic development of students. The institution takes special efforts to inculcate a conducive green environment for teaching-learning process. The 'Green Campus' culture is imbibed among the students so that it becomes an integral part of their personal growth. Following are the initiatives taken by the institution for greening the campus:

1.RESTRICTED ENTRY OF AUTOMOBILES

The ambience of the College is calm, spacious with good amenities and utilities in a visually appealing landscape. To maintain a pollution free environment, automobiles are restricted inside the campus beyond the parking area. Vehicle parking space is provided at the main entrance of the college campus. The building in the campus is designed in such a way that the area surrounding the classrooms and playgrounds have a sound free and pollution free environment for students. It is ensured that all the vehicles used by staff and students for daily commute are PUC certified. Staff and students are encouraged to take public transport, bicycle, walk, and carpool to campus.

2. USE OF BICYCLES/ BATTERY-POWERED VEHICLES

The location of the college is such an advantage that it has connectivity with the city buses which the staff and students use on daily basis. Locational advantage again takes credit to this that many staff members and students who are staying in the nearby area are encouraged to use bicycles or battery-operated vehicles. Around 20% students and 3% of staff members use bicycles for daily commute. To spread awareness and maintain habit, ‘Bicycle Day’ is celebrated in the campus.

3. PEDESTRIAN-FRIENDLY PATHWAYS

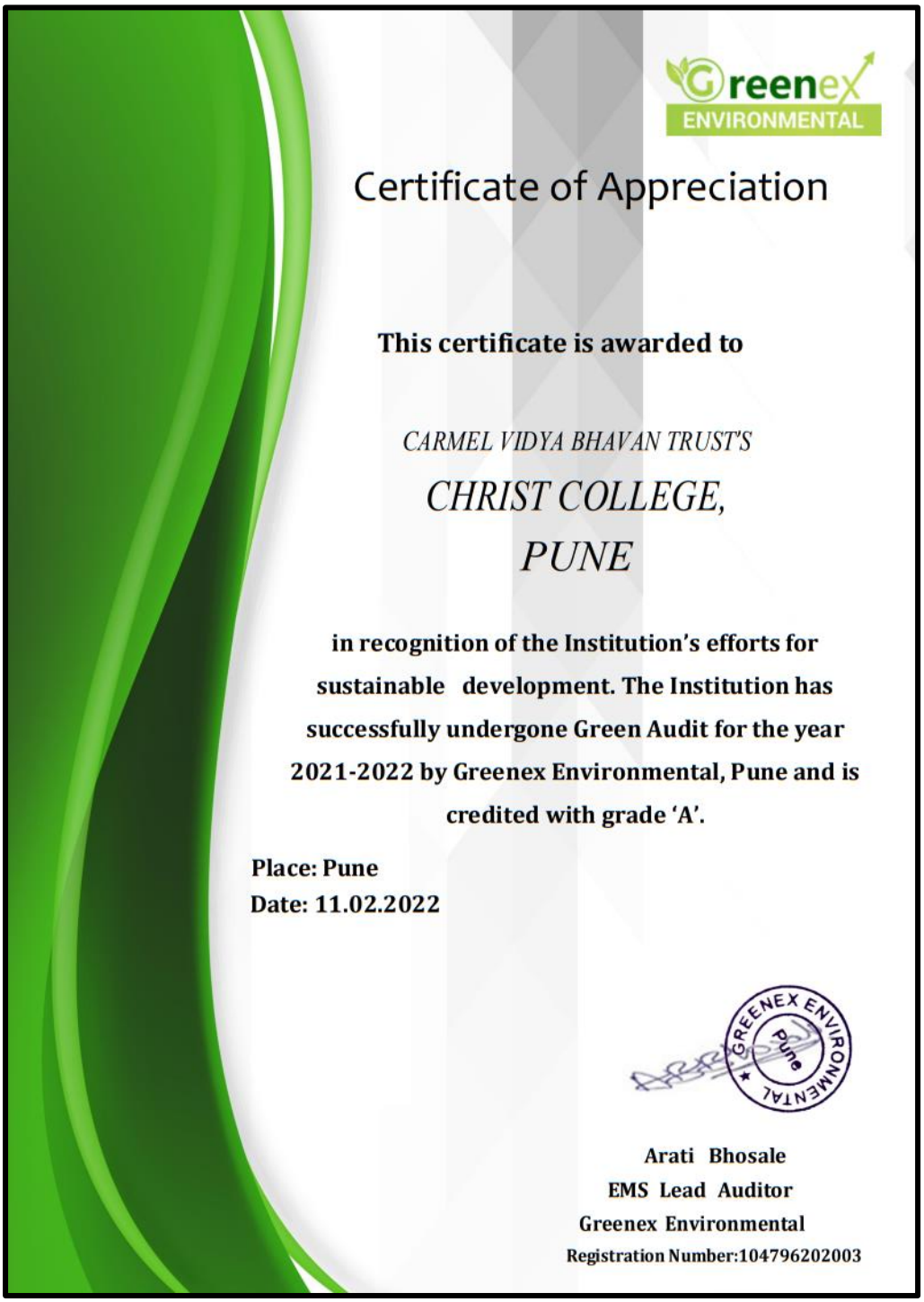
To facilitate safe and easy movement around the lush green campus, pedestrian-friendly pathways are made which give wide walking area inside the campus.

4. LANDSCAPING

College campus is having worth seeing lush green canopy of trees and plants which make the environment pollution free and provides soothing and serene environment for making environment healthy climate for studying and learning. The institute works to develop an eco-friendly, sustainable campus and to disseminate the concept of eco-friendly culture. The campus has a green landscape where students can freely sit and have academic discussions. This environment is maintained so that students feel one with nature. 60% of the 5 acres land is covered by more than 400 trees of varied varieties. The college has developed a garden with many seasonal, perennial, and medicinal plants. Keeping

a complete balance of flora and fauna for a green ecosystem domestic animals like rabbits, guinea pigs, hens, black chickens, goats, fishes, and various varieties of birds are reared. Organic vegetables and fruits are cultivated and distributed to the neighboring communities free of cost. The students are encouraged to actively participate in activities like plantation, cleanliness drives, awareness rallies, campaigns, exhibitions, and lectures.

GREEN AUDIT, ENVIRONMENT AND ENERGY AUDITS



Greenex ENVIRONMENTAL

Certificate of Appreciation

This certificate is awarded to

CARMEL VIDYA BHAVAN TRUST'S

CHRIST COLLEGE,

PUNE

in recognition of the Institution's efforts for sustainable development. The Institution has successfully undergone Green Audit for the year 2021-2022 by Greenex Environmental, Pune and is credited with grade 'A'.

Place: Pune
Date: 11.02.2022

GREENEX ENVIRONMENTAL Pune

Arati Bhosale
EMS Lead Auditor
Greenex Environmental
Registration Number:104796202003

Greenex ENVIRONMENTAL

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ENERTEK
SOLUTIONS
India Private Limited

ENERGY AUDIT CERTIFICATE

The study is conducted as per Bureau of Energy Efficiency Standards initiated in the capacity of an Accredited & Certified Energy Audit Professional

It is awarded for **2022-2023** to the Esteemed Institution

"CHRIST COLLEGE, PUNE"

26/4 A, Off Pune Nagar Road, Opposite Weikfield IT Citi Info Park, 26/4A, Nagar Rd, Ramwadi, Wadgaon Sheri, Pune, Maharashtra 411014 (On the basis of the actual site visit at college)

As part of the Institution's initiatives for a Healthy & Sustainable College the audit was conducted.
We appreciate the immense efforts taken by Staff and students towards the Energy Management and Conservation.

ENERTEK SOLUTIONS INDIA PRIVATE LIMITED

Mr. Harshad Batule
BEE-Certified Energy Auditor
(CEA – 34724)
harshad@enertecsolutions.com
Director
ENERTEK SOLUTIONS INDIA PVT LTD
615-B, Nana Peth, Nr. Parasi Agyari,
Pune – 411 002

BEYOND THE CAMPUS ENVIRONMENTAL PROMOTIONAL ACTIVITIES

Pledges taken by NSS Volunteers

NSS volunteers took pledge for saving environment in various ways through the government website- Unified Online Pledge Platform of MyGov. https://pledge.mygov.in. The purpose of the #Epledge is to motivate each citizen to take a conscious and collaborative effort towards the betterment of the environment.









PLANTATION DRIVES



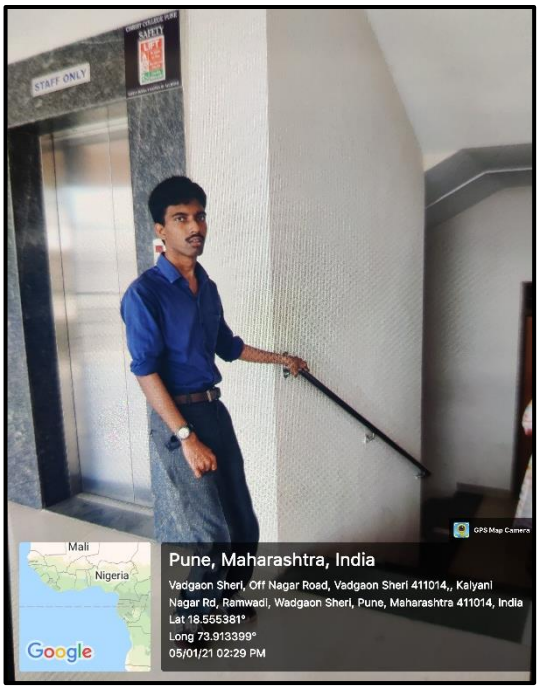


5. DISABLED-FRIENDLY BARRIER FREE ENVIRONMENT

1.BUILT ENVIRONMENT WITH RAMPS/LIFTS FOR EASY ACCESS TO CLASSROOMS

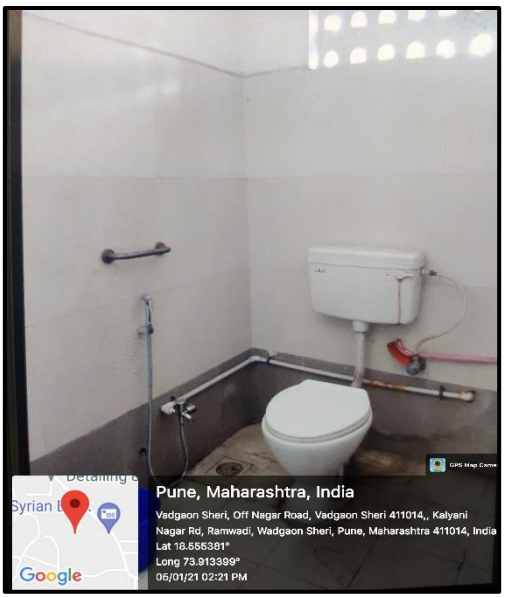
The campus is equipped with Ramp-Rails and Lift in addition to staircases in the campus. The rails are carefully designed as per specifications to be used by the differently-abled people.

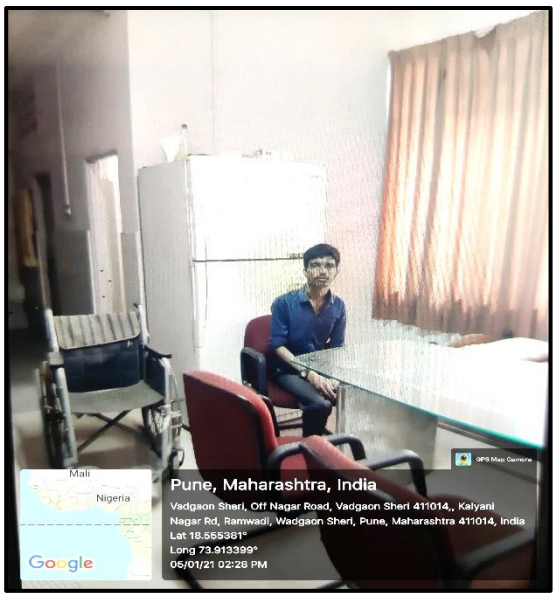




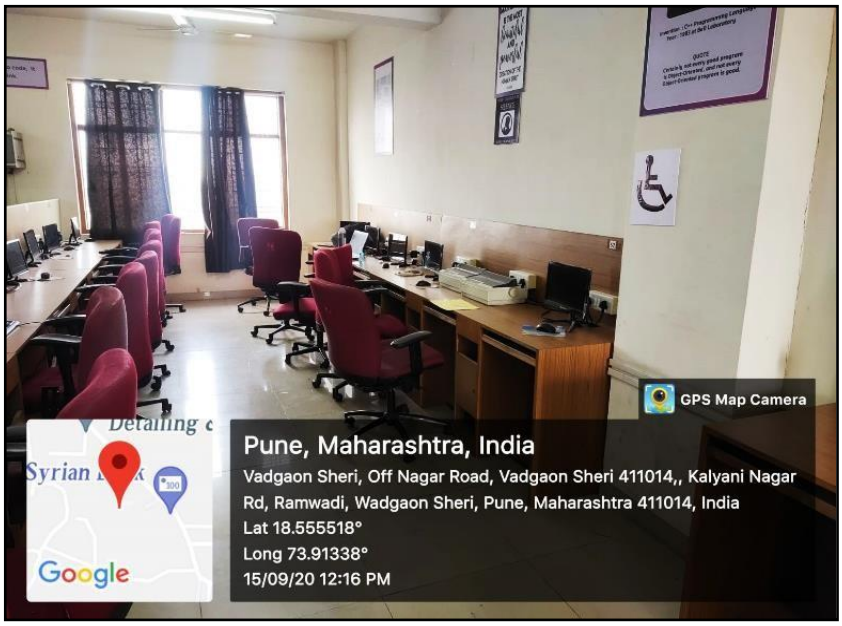
2.DISABLED-FRIENDLY WASHROOMS AND REST ROOM

The campus has accessible restroom with the fixtures and fittings that are comfortable and convenient to the differently-abled people. The washroom has nonslip floor and grab bars that provide easy access to the differently-abled people.





3. SIGNAGE, LIGHTS, DISPLAY BOARDS AND SIGNPOSTS



4. ASSISTIVE TECHNOLOGY AND FACILITIES FOR PERSONS WITH DISABILITIES (DIVYANGJAN)
ACCESSIBLE WEBSITE, SCREEN-READING SOFTWARE, MECHANIZED EQUIPMENT

The college library has installed the NVDA NV Access software for differently-abled students. NonVisual Desktop Access is a free and open-source, portable screen reader for Microsoft Windows. ***NVDA*** allows blind and vision impaired people to access and interact with the ***Windows*** operating system and many third party applications.

Alexa- virtual assistant technology is also used in the library.

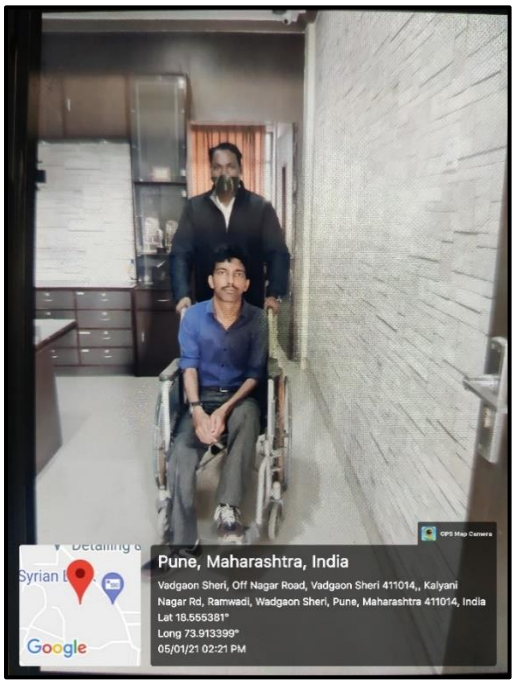


5. PROVISION FOR ENQUIRY AND INFORMATION: HUMAN ASSISTANCE, READER, SCRIBE, SOFT COPIES OF READING MATERIAL, SCREEN READING

- The college provides Reception Counter for information to the differently-abled people.



- **Wheelchair is provided with human assistance**

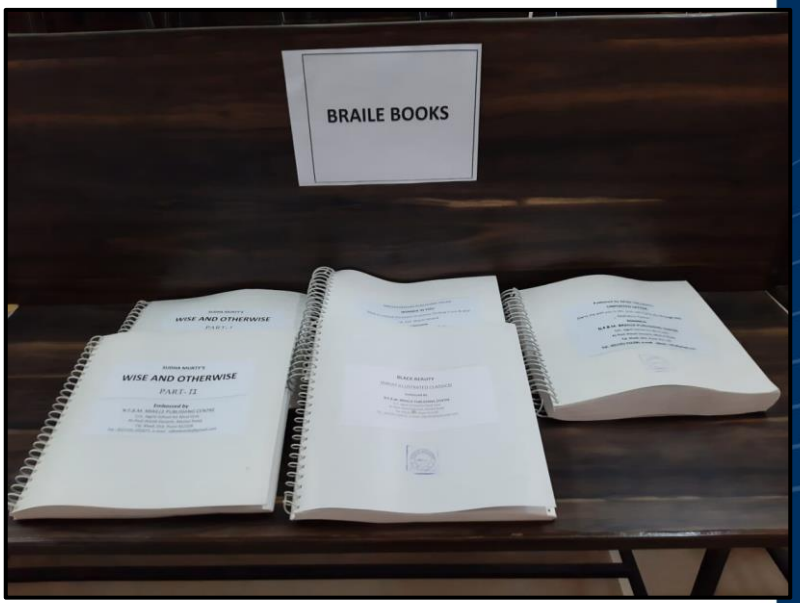


- Facility of providing Scribes during examination



- Braille books are provided in library





- Separate sitting area in lab and library to provide safety and comfort

